Exercise J - A

1. A value of b for which equations $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have one root in common is : **[JEE(Adv)-2011]**

b के किस मान के लिये समीकरणों $x^2 + bx - 1 = 0$ तथा $x^2 + x + b = 0$ का एक मूल उभयनिष्ठ होगा : **[JEE(Adv)-2011]**

(A) $-\sqrt{2}$ (B) $-i\sqrt{3}$ (C) $i\sqrt{5}$ (D) $\sqrt{2}$

Ans. B

Sol. $$\begin{array}{l} x^2 + bx - 1 = 0 \\ x^2 + x + b = 0 \end{array} \Rightarrow \frac{x^2}{b^2+1} = \frac{x}{-1-b} = \frac{1}{1-b} \Rightarrow x = \frac{b^2+1}{-(b+1)} = \frac{-(b+1)}{1-b} \Rightarrow (b^2+1)(1-b) = (b+1)^2$$

$\Rightarrow \quad b^2 - b^3 + 1 - b = b^2 + 2b + 1 \Rightarrow b^3 + 3b = 0 \Rightarrow b = 0 ; b^2 = -3 \Rightarrow b = 0, \pm\sqrt{3}\,i$

2. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation $p(p(x)) = 0$ has : **[JEE(Adv)-2014]**

(A) only purely imaginary roots (B) all real roots

(C) two real and two purely imaginary roots (D) neither real nor purely imaginary roots

समीकरण $p(x) = 0$, सभी गुणांक वास्तविक है, के शुद्ध काल्पनिक मूल है तो समीकरण $p(p(x)) = 0$ के : **[JEE(Adv)-2014]**

(A) केवल शुद्ध काल्पनिक मूल होगें (B) सभी मूल वास्तविक होगे

(C) दो वास्तविक तथा दो काल्पनिक मूल होगे (D) ना तो वास्तविक ना ही शुद्ध काल्पनिक मूल होगे

Ans. D

Sol. $p(x)$ will be of the form $ax^2 + c$. Since it has purely imaginary roots only.

Since $p(x)$ is zero at imaginary values while $ax^2 + c$ takes real value only at real 'x', no root is real.

Also $p(p(x)) = 0 \Rightarrow p(x)$ is purely imaginary $\Rightarrow ax^2 + c =$ purely imaginary

Hence x can not be purely imaginary since x^2 will be negative in that case and $ax^2 + c$ will be real.

Thus .(D) is correct.

3. Let S be the set of all non-zero numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is (are) a subset (s) of S? **[JEE(Adv)-2015]**

माना कि S उन सभी शून्येतर (non-zero) वास्तविक संख्याओं α का समुच्चय (set) है जिनके लिए द्विघाती समीकरण $\alpha x^2 - x + \alpha = 0$ के दो विभिन्न वास्तविक मूल x_1 और x_2 असमिका $|x_1 - x_2| < 1$ को सन्तुष्ट करते हैं। निम्नलिखित अन्तरालों में से कौनसा (से) समुच्चय S के उपसमुच्चय है (हैं)? **[JEE(Adv)-2015]**

(A) $\left(-\frac{1}{2},-\frac{1}{\sqrt{5}}\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)$ (C) $\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(\frac{1}{\sqrt{5}},\frac{1}{2}\right)$

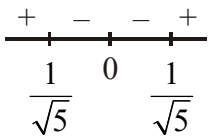
Ans. AD

Sol. $(x_1+x_2)^2-4x_1x_2<1$

$$\frac{1}{\alpha^2}-4<1$$

$$\Rightarrow \quad 5-\frac{1}{\alpha^2}>0$$

$$\frac{5\alpha^2-1}{\alpha^2}>0$$



$$\alpha\in\left(-\infty,-\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right) \qquad \text{_____(1)}$$

$D > 0$

$1 - 4\alpha^2 > 0$

$$\alpha\in\left(-\frac{1}{2},\frac{1}{2}\right) \qquad \text{_____(2)}$$

(1) and (2)

$$\alpha\in\left(-\frac{1}{2},\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\frac{1}{\sqrt{2}}\right)$$

Paragraph for Question nos. 4 to 5

Let p, q be integers and let α, β be the roots of the equation, $x^2 - x - 1 = 0$, where $\alpha \neq \beta$. For $n = 0,1, 2,.....$, let $a_n = p\alpha^n + q\beta^n$.

FACT : if a and b are rational numbers and $a + b\sqrt{5} = 0$, then $a = 0 = b$. **[JEE(Adv)-2017]**

माना कि p, q पूर्णांक है एवं α, β समीकरण $x^2 - x - 1 = 0$ के मूल हैं, जहाँ $\alpha \neq \beta$ है। $n = 0,1, 2,.....$, के लिये माना कि $a_n = p\alpha^n + q\beta^n$.

तथ्यः यदि a तथा b परिमेय संख्यायें हैं एवं $a + b\sqrt{5} = 0$ है, तब $a = 0 = b$. **[JEE(Adv)-2017]**

4. If $a_4 = 28$, then $p + 2q =$

यदि $a_4 = 28$ है, तब $p + 2q =$

(A) 12 (B) 21 (C) 14 (D) 7

Ans. A

Sol. $a_{n+2} = a_{n+1} + a_n$

$a_4 = a_3 + a_2 = 3a_1 + 2a_0 = 3p\alpha + 3q\beta + 2(p + q)$

As $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$, we get

$$a_4 = 3p\left(\frac{1+\sqrt{5}}{2}\right) + 3q\left(\frac{1-\sqrt{5}}{2}\right) + 2p + 2q = 28$$

$$\left(\frac{3p}{2} + \frac{3q}{2} + 2p + 2q - 28\right) = 0 \text{.....(i)}$$

and $\frac{3p}{2} - \frac{3q}{2} = 0$.....(ii)

$\Rightarrow$ $p = q$ (from (ii))

$\Rightarrow$ $7p = 28$ (from (i) and (ii))

$\Rightarrow$ $p = 4$

$\Rightarrow$ $q = 4$

$\Rightarrow$ $p + 2q = 12$

5. $a_{12} =$

(A) $a_{11} + 2a_{10}$ (B) $a_{11} + a_{10}$ (C) $a_{11} - a_{10}$ (D) $2a_{11} + a_{10}$

Ans. B

Sol. As α and β are roots of equation $x^2 - x - 1 = 0$, we get :

$\alpha^2 - \alpha - 1 = 0 \quad \Rightarrow \quad \alpha^2 = \alpha + 1$

$\beta^2 - \beta - 1 = 0 \quad \Rightarrow \quad \beta^2 = \beta + 1$

$\therefore \quad a_{11} + a_{10} = p\alpha^{11} + q\beta^{11} + p\alpha^{10} + q\beta^{10}$

$= p\alpha^{10}(\alpha + 1) + q\beta^{10}(\beta + 1)$

$= p\alpha^{10} \times \alpha^2 + q\beta^{10} \times \beta^2$

$= p\alpha^{12} + q\beta^{12}$

$= a_{12}$

6. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is : **[JEE(Adv)-2020]**

मान लीजिये कि a, b द्विघातीय बहुपद $x^2 + 20x - 2020$ के भिन्न वास्तविक मूलों को दर्शाते हैं, एवं मान लीजिये कि c, d द्विघातीय बहुपद $x^2 - 20x + 2020$ के भिन्न समिश्र मूलों को दर्शाते हैं। तब $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ का मान है : **[JEE(Adv)-2020]**

(A) 0 (B) 8000 (C) 8080 (D) 16000

Ans. D

Sol. $x^2 + 20x - 2020 = 0$ a, b are its roots

$x^2 - 20x + 2020 = 0$ c,d are its roots

$a + b = -20$ $ab = -2020$

$c + d = 20$ $cd = 2020$

$E = ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$

$= a^2c - ac^2 + a^2d - ad^2 + b^2c + b^2d - bd^2$

$= a^2(c + d) + b^2(c + d) - c^2(a + b) - d^2(a + b)$

$= (a^2 + b^2)(c + d) - (a + b)(c^2 + d^2)$

$= ((a + b)^2 - 2ab)(c + d) - (a + b)((c + d)^2 - 2cd)$

$= (400 + 4040)(20) - (-20)(400 - 4040)$

$= 88800 - 72800 = 16000$

7. Let R^2 denote $R \times R$. Let $S = \{(a, b, c) : a, b, c \in R$ and $ax^2 + 2bxy + cy^2 > 0$ for all $(x, y) \in R^2 - \{(0,0)\}\}$. Then which of the following statements is(are) TRUE ? **[JEE(Adv.)2024]**

(A) $\left(2, \frac{7}{2}, 6\right) \in S$

(B) If $\left(3, b, \frac{1}{12}\right) \in S$, then $|2b| < 1$.

(C) For any given $(a, b, c) \in S$, the system of linear equations
$ax + by = 1$
$bx + cy = -1$
has a unique solution.

(D) For any given $(a, b, c) \in S$, the system of linear equations
$(a + 1)x + by = 0$
$bx + (c + 1)y = 0$
has a unique solution.

माना कि R^2, $R \times R$ को दर्शाता है। माना कि $S = \{(a, b, c) : a, b, c \in R$ और सभी $(x, y) \in R^2 - \{(0,0)\}$ के लिए $ax^2 + 2bxy + cy^2 > 0\}$ है। तब निम्नलिखित कथनों में से कौनसा(से) सत्य है(हैं) ? **[JEE(Adv.)2024]**

(A) $\left(2, \frac{7}{2}, 6\right) \in S$

(B) यदि $\left(3, b, \frac{1}{12}\right) \in S$, तब $|2b| < 1$ है।

(C) किसी दिये गये $(a, b, c) \in S$ के लिए, रैखिक समीकरणों के निकाय

$ax + by = 1$

$bx + cy = -1$

का एक अद्वितीय हल है।

(D) किसी दिये गए $(a, b, c) \in S$ के लिए, रैखिक समीकरणों के निकाय

$(a + 1)x + by = 0$

$bx + (c + 1)y = 0$

का एक अद्वितीय हल है।

Ans. BCD

Sol. $ax^2 + 2bxy + cy^2 > 0$

$\Rightarrow a\left(\frac{x}{y}\right)^2 + 2b\left(\frac{x}{y}\right) + c > 0$ $\quad$ $c\left(\frac{y}{x}\right)^2 + 2b\left(\frac{y}{x}\right) + a > 0$

$\Rightarrow a > 0$ and $D < 0$ $\quad$ $c > 0$ and $D < 0$

$b^2 < ac$ and $a > 0$ and $c > 0$

(A) $\frac{49}{4} < 12$ incorrect

(B) $b^2 < \frac{3}{12} \Rightarrow 4b^2 < 1 \quad \Rightarrow |2b| < 1$ correct

(C) for unique solution $\frac{a}{b} \neq \frac{b}{c} \Rightarrow b^2 \neq ac$ correct

(D) $\frac{a+1}{b} \neq \frac{b}{c+1} \Rightarrow \underbrace{b^2 - ac}_{(-)ve} \neq \underbrace{a + c + 1}_{(+)ve}$ correct

8. Let $f(x) = x^4 + ax^3 + bx^2 + c$ be a polynomial with real coefficients such that $f(1) = -9$. Suppose that $i\sqrt{3}$ is a root of the equation $4x^3 + 3ax^2 + 2bx = 0$, where $i = \sqrt{-1}$. If α_1, α_2, α_3 and α_4 are all the roots of the equation $f(x) = 0$, then $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ is equal to _____. **[JEE(Adv.)2024]**

माना कि $f(x) = x^4 + ax^3 + bx^2 + c$ वास्तविक गुणांकों वाला एक ऐसा बहुपद है कि $f(1) = -9$ है। मान लीजिये कि $i\sqrt{3}$, समीकरण $4x^3 + 3ax^2 + 2bx = 0$ का एक मूल है, जहाँ $i = \sqrt{-1}$ है। यदि α_1, α_2, α_3 और α_4, समीकरण $f(x) = 0$ के सभी मूल हैं, तब $|\alpha_1|^2 + |\alpha_2|^2 + |\alpha_3|^2 + |\alpha_4|^2$ का मान _____ है। **[JEE(Adv.)2024]**

Ans. 20

Sol. $f(1) = -9$

$a + b + c = -10$

$4x^3 + 3ax^2 + 2bx = 0$

$x\left(4x^2 + 3ax + 2b\right) = 0$

$\Rightarrow \left(4x^2 + 3ax + 2b\right) = 0 \begin{matrix} \nearrow i\sqrt{3} \\ \searrow -i\sqrt{3} \end{matrix}$

$a = 0,\ \frac{2b}{4} = 3 \Rightarrow b = 6$

$c = -10 - a - b = -16$

$f(x) = x^4 + 6x^2 - 16 = 0$

$(x^2 + 8)(x^2 - 2) = 0$

$x = \pm\sqrt{2}\ ,\ \pm 2\sqrt{2}i$

$\sum|\alpha|^2 = 20$

9. Let $a = 3\sqrt{2}$ and $b = \frac{1}{5^{\frac{1}{6}}\sqrt{6}}$. If x, y ∈ R such that $3x + 2y = \log_a (18)^{\frac{5}{4}}$ and $2x - y = \log_b\left(\sqrt{1080}\right)$, then $4x + 5y$ is equal to _____. **[JEE(Adv.)2024]**

माना कि $a = 3\sqrt{2}$ और $b = \frac{1}{5^{\frac{1}{6}}\sqrt{6}}$ हैं। यदि x, y ∈ R इस प्रकार हैं कि $3x + 2y = \log_a (18)^{\frac{5}{4}}$ और $2x - y = \log_b\left(\sqrt{1080}\right)$, तब $4x + 5y$ बराबर _____ है। **[JEE(Adv.)2024]**

Ans. 8

Sol. $\log_{3\sqrt{2}}\left(3\sqrt{2}\right)^{\frac{5}{2}} = \frac{5}{2} = 3x + 2y \quad \Rightarrow 3x + 2y = \frac{5}{2} \quad _____(1)$

$2x - y = \log_{\left(5^{\frac{1}{6}}\sqrt{6}\right)^{-1}} \sqrt{5^1 6^3} = -3 \quad \Rightarrow 4x - 2y = -6 \quad _____(2)$

From equ. (1) and (2), $x = -\frac{1}{2}, y = 2$

$4x + 5y = -2 + 10 = 8$